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R Notebook (FEA): Facial expression analysis thresholding and aggregation

Learn how to calculate whether a facial expression was present or absent with the FEA thresholding and aggregation R Notebooks developed for the three supported facial coding algorithms from Affectiva AFFDEX, Emotient FACET, and Realeyes.

Background

The raw signal from the facial coding algorithm is a confidence score indicating how sure the algorithm is at a given point in time that it detected a given facial expression.

Through data analysis with the R Notebook, iMotions can filter out those periods of time where the algorithm was unsure about the classification of a facial expression. By keeping only those periods of time where the algorithm classified a facial expression with high confidence, one can classify whether a facial expression was present (or absent).

In iMotions, this is done by applying a threshold used to binarize the raw signal.

Through this thresholding procedure, iMotions can indicate when in time a facial expression started (onset event) and ended (offset event).

iMotions can also classify whether a facial expression was present for a short or long period of time (compared against a user defined time threshold).

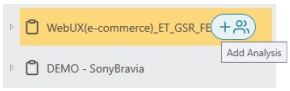
By aggregating the thresholded signals from many respondents, iMotions can calculate how many respondents showed a given facial expression at any point in time.

For data analysis, iMotions can calculate summary metrics allowing the user to export aggregated emotion metrics which is a way to reduce the massive amount of information from the raw data export (Sensor Data Export) into a summary table that can be used to compare emotion scores between stimuli, groups etc.

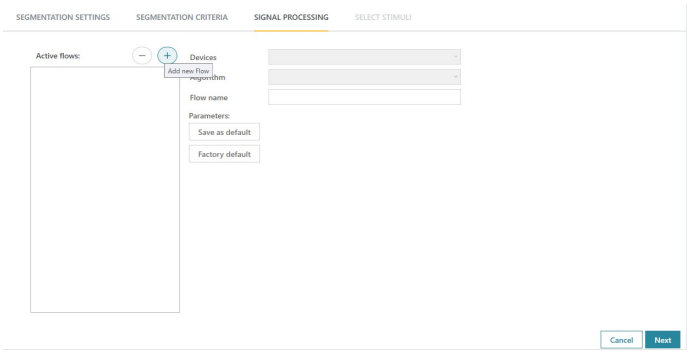
How to run the processing

Please note that we recommend post processing all respondent videos before starting an analysis of the facial expression data

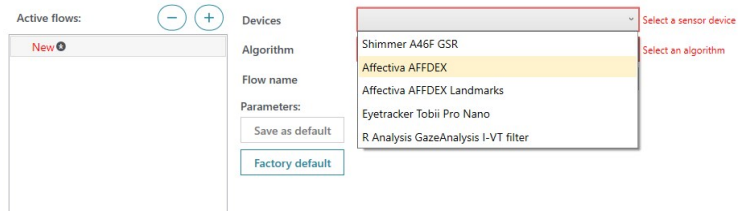
1. Create an Analysis by right-clicking on the study and selecting "add analysis" or by clicking on the oval button



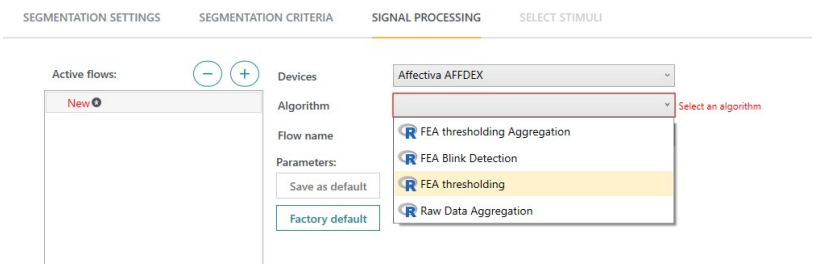
2. Under Signal processing, click the "+" button to add a new Active Flow



3. Under "Device", choose the facial coding algorithm you used



4. Mandatory: Under "Algorithm", choose "FEA thresholding"



5. Decide on the threshold values you want to apply. The output will reflect the number of data points and percent of time above the threshold value for each category below.
You can either batch apply a threshold value for the emotion channels and facial expressions, or choose to set individual thresholds for each channel individually.

It is recommended that you explore and normalize your own value dataset but as a general rule of thumb, 50% confidence represents a moderately strong display of facial response. 75% or higher means there was high confidence in classifying the facial expression and 25% or lower means that only 1 in 4 manual coders has classified the facial expression as present.

You can also choose if you wish to classify every present facial expression into short or long.

Default Threshold parameters for Affectiva

- Emotion: 50
- Valence: + 50 (Positive Valence) and - 50 (Negative Valence)

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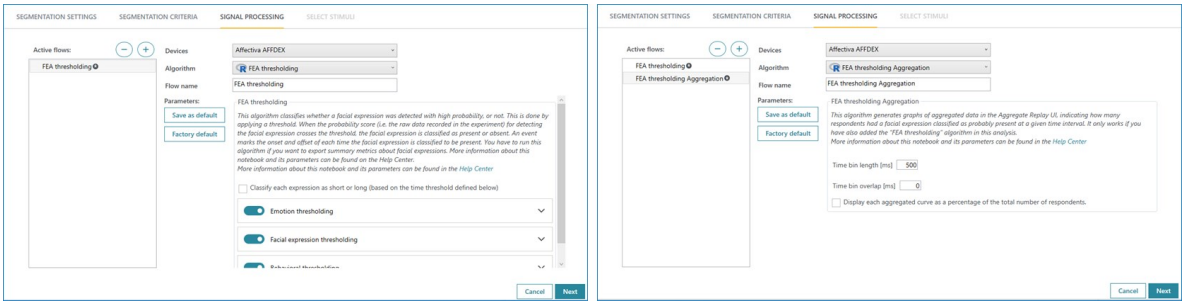
- Action Units: 50
- Behavioral signal thresholds (recommended by Affectiva)
  - Right (deg): Yaw greater than +10
  - Left (deg): Yaw less than -10
  - Up (deg): Pitch greater than 0
  - Down (deg): Pitch less than -20
  - Right tilt (deg): Roll less than -10
  - Left tilt (deg): Roll greater than +10
  - Forward (px): Interocular distance greater than 100 px
  - Backward (px): Interocular distance less than 100 px

6. Optional: Under Algorithm, choose "FEA thresholding Aggregation"

This step is only required if you wish to view graphs for static stimuli or videos in the Aggregate Replay UI in iMotions. If you only wish to process data and export statistics, this step is not required.

The parameters give you the ability to change the bin length and bin overlap that the aggregation is relied on. You can also choose to visualize the aggregated curves in numbers of respondents or in percentage.

7. Complete the analysis creation.



Signal Processing Flows with 'FEA Thresholding' (left) and 'FEA Thresholding Aggregation' (right)

Method

Facial expressions are classified as being probably present or absent based on a binarization of the raw data collected in the experiment. A threshold is applied, indicating when the confidence score of a given facial expression (i.e. the raw data) exceeds or surpasses the chosen threshold, hence classifying the facial expression as present or absent.

If raw data is missing, the classifier handles it the following way:

- If data right before and after the gap is classified as absent, also the time with data loss is classified as absent
- If data right before and after the gap is classified as present, also the time with data loss is classified as present
- If data is classified as present or absent before the gap, but as the opposite after the data gap, the time with data loss is classified as absent

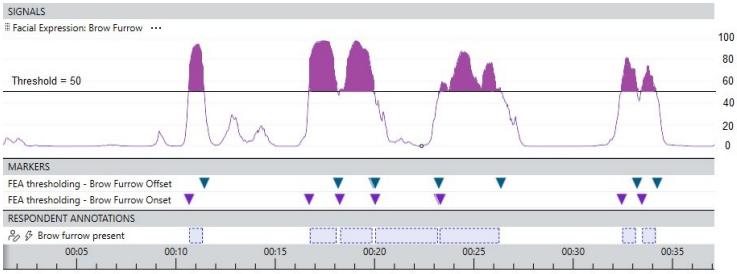
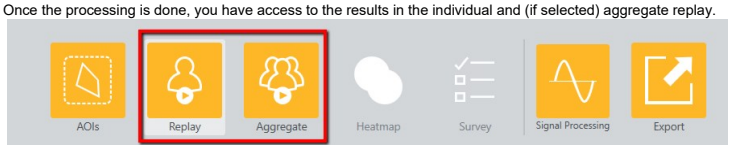


Illustration of how applying a threshold (here: 50) classifies a facial expression (here: brow furrow) into present (highlighted) and absent

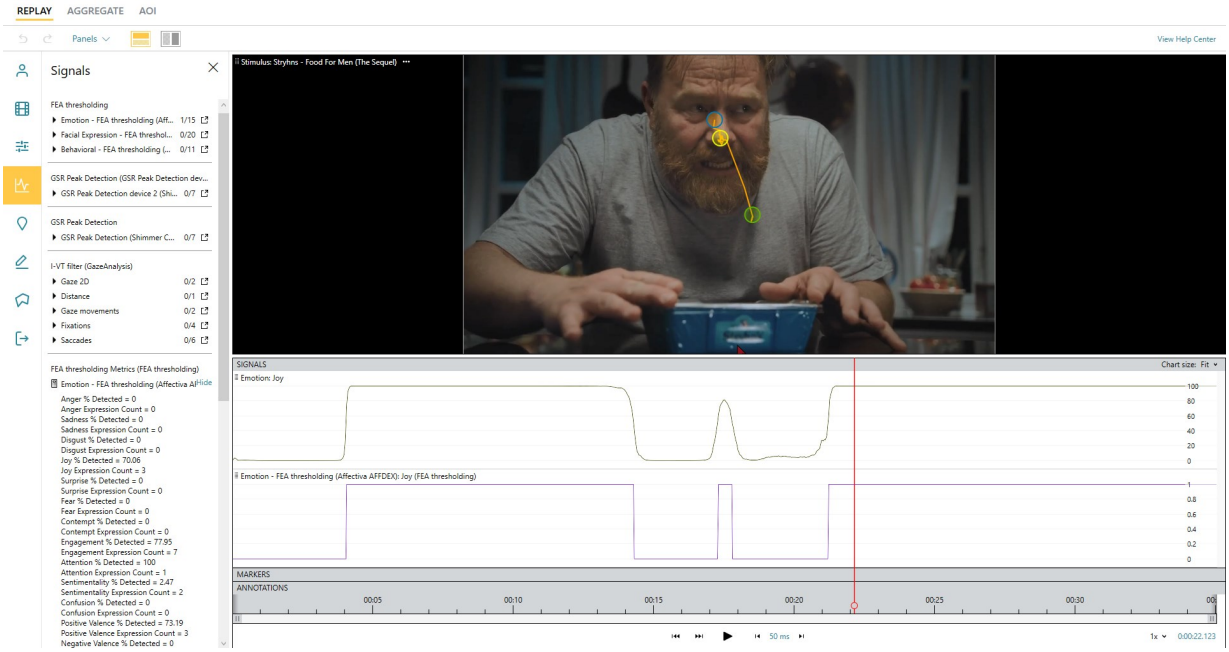
Results

Once the processing is done, you have access to the results in the individual and (if selected) aggregate replay.



Results in the Individual Replay

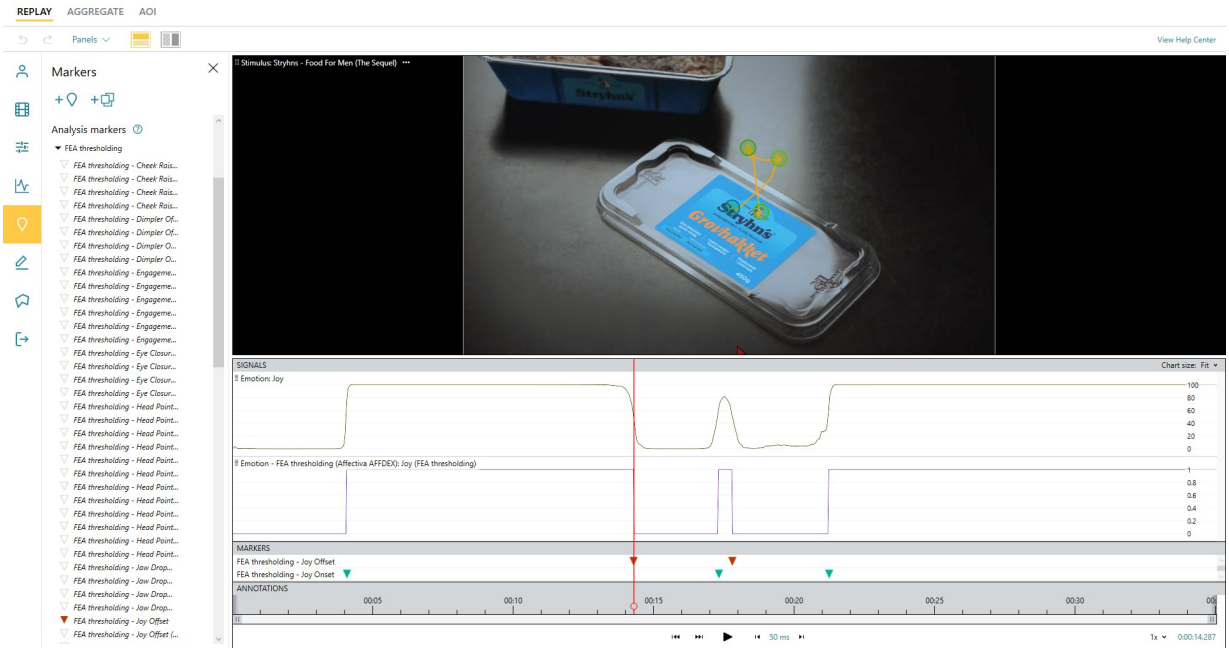
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Data in the Individual Replay

In addition to the raw data graphs for facial expression analysis, an extra set of graphs can be selected from the Signals: Under "FEA thresholding", you find the binarized signals indicating when a facial expression was classified as present (1) or absent (0).

FEA thresholding Metrics are given for every facial expression, indicating the percentage (%) of time out of the entire stimulus duration for which the facial expression is present as well as the amount of segments with the facial expression present (i.e. how often it was expressed on the stimulus).



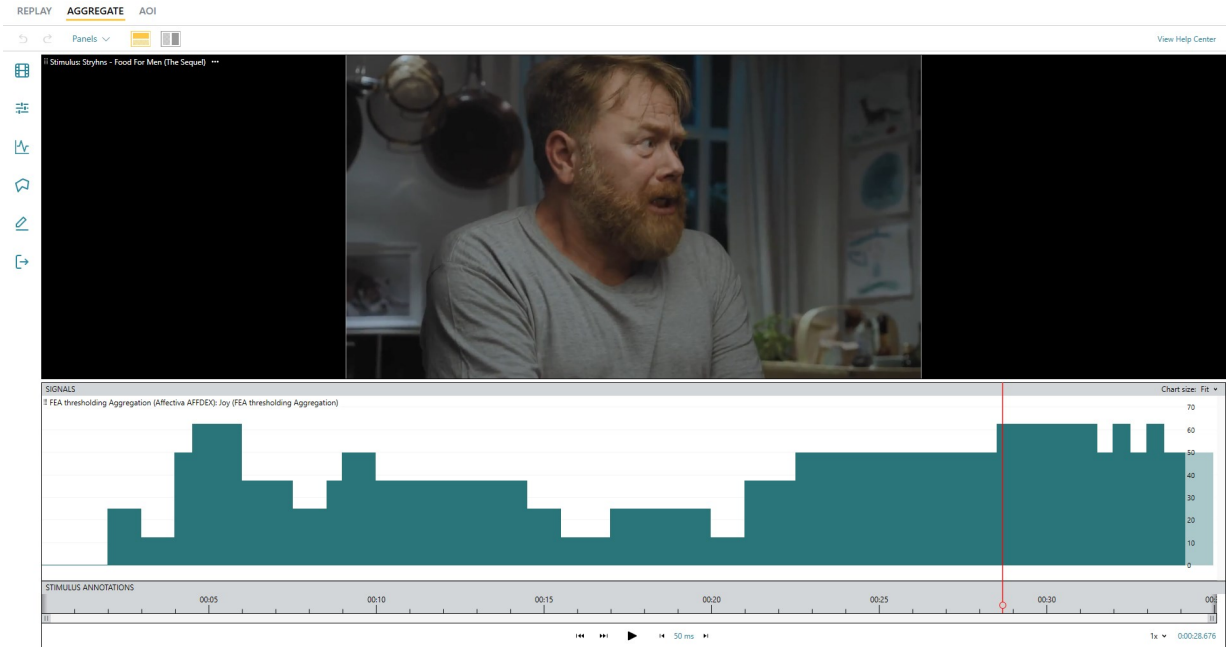
Events in the Individual Replay

In the Markers menu, a list of all onset and offset events for all facial expressions can be retrieved.

Please note that these events can be used to place automated Annotations.

Results in the Aggregated Replay

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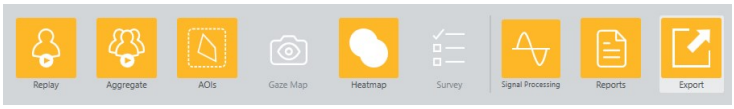
Graph in the Aggregated Replay indicating the percentage of respondents showing Joy at the same time

If you chose to run the FEA thresholding Aggregation, you can choose to see graphs in the Aggregate replay indicating how many respondents showed a selected facial expression within a time window.

Export options

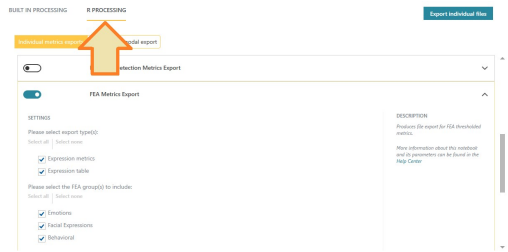
In order to run statistical data analysis, you can export the results. iMotions offers you different export options.

You can find the Export button in the bottom menu when in the analysis level.



FEA Metrics Export

To export summary metrics, go to Export > R Processing > FEA Metrics Export



Access the FEA Metrics exports through *Export > R Processing*

You can choose if you wish to export metrics for all available facial expressions, or only for certain facial expressions.

You can also choose to include Annotation markers in the facial expressions table.

Expression metrics provide a summary of how often and how long a facial expression was present for each respondent and each time interval (Stimulus, Annotation, Scene).

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#METADATA																
#Channel identi	Info_Resp	Info_Resp	Info_Resp	Info_Resp	Info_Inter	Info_Inter	Info_Inter	Info_Inter	Info_Inter	Info_Inter	FEA_Thresholded	FEA_Thresholded	FEA_Thresholded	FEA_Thresholded	FEA_Thresholded	FEA_Thresholded
#Group	Responde	Responde	Responde	Responde	Interval In	Interval In	Interval In	Interval In	Interval In	Interval In	FEA threshold me	FEA threshold me	FEA threshold me	FEA threshold me	FEA threshold me	FEA threshold me
#Unit							Millisecor	Milliseconds						Percentage		
#Short unit							ms	ms						%		
#Description	Name of the respondent.	Gender of the respondent.	Age of the respondent.	Group of the respondent.	Type of the interval that the metrics refers to.	Name of the interval.	of the interval (relative to recording start).	Duration of the interval.	Name of the parent.	Comment that was added to the interval.	Count of all data points (video frames) in the interval.	Threshold defined by the user for Joy.	Count of data points (video frames) for which Joy was classified.	Percentage of the interval's duration during which Joy was classified.	Count of long Joy expressions.	Count of short Joy expressions.
#Device											Affectiva AFFDEX	Affectiva AFFDEX	Affectiva AFFDEX	Affectiva AFFDEX	Affectiva AFFDEX	Affectiva AFFDEX
#AlgorithmType											FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding
#Flow Name											FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding	FEA thresholding
#DATA																
Study Name	Responde	Gender	Age	Group	Type	Label	Start (ms)	Duration (	Parent Sti	Comment	Count Frames	Joy Threshold	Joy Frames >= Thr	Joy Time Percenta	Joy Long Expressi	Joy Short Express
Media (90s Videc	Anonymot	FEMALE	0	Default	Stimulus	instructio	711	9648			288	50	162	56.25	1	2
Media (90s Videc	Anonymot	FEMALE	0	Default	Stimulus	90secBase	10503	89803			2691	50	679	25.23	5	2
Media (90s Videc	Anonymot	FEMALE	0	Default	Annotatio	Coral reef	10503	19846	90secBaseline		597	50	199	33.33	2	1
Media (90s Videc	Anonymot	FEMALE	0	Default	Annotatio	Roof	30349	31764	90secBaseline		945	50	0	0	0	0
Media (90s Videc	Anonymot	FEMALE	0	Default	Annotatio	Seaside	62113	21090	90secBaseline		635	50	245	38.58	1	0
Media (90s Videc	Anonymot	FEMALE	0	Default	Annotatio	Baby	83203	17095	90secBaseline		514	50	235	45.72	2	1

The facial expressions metrics export

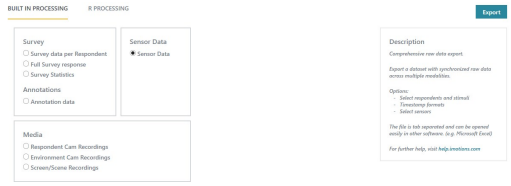
Expression table lists all instances of a facial expression being present along with the onset and offset time of this segment, its duration, information on whether the expression was classified as short or long, as well as a running number counting the instances.

#METADATA																			
#Channel ide	Info_Resp	Info_Resp	Info_Resp	Info_Resp	Info_Inter	Info_Inter	Info_Inter	FEA_Expre	FEA_Threshold	FEA_Expression	FEA_Expression	FEA_Expressior	FEA_Expressior	FEA_Expressior	FEA_TruncationType	FEA_DurationClassificatior	FEA_TimeThreshold	FEA_Device
#Unit					Millisecor	Milliseconds						Milliseconds	Milliseconds	Milliseconds			Milliseconds	
#Short unit					ms	ms						ms	ms	ms			ms	
#Description	Name of the respondent.	Gender of the respondent.	Age of the respondent.	Group of the respondent.	Name of the stimulus.	Start time of the stimulus (relative to recording start).	Duration of the stimulus.	Facial expressio n classifie d.	Threshold defined by the user for the respective expression.	Expression instance number, counting from start of the recording.	Expression instance number, counting from start of the stimulus.	Onset time of the facial expression (relative to recording start).	Offset time of the facial expression (relative to recording start).	Duration of the facial expression.	Whole: The facial expression started and ended within the interval; Partial: The facial expression started or ended outside of the interval. Data was truncated to start or end within the interval.	Whether the facial expression was longer or shorter than the indicated time threshold, based on the entire duration of the facial expression.	Time threshold defined by the user to classify long vs. short expressions.	Algorithm used for facial expression analysis.
#DATA																		
Study Name	Responde	Responde	Responde	Responde	Stimulus I	Stimulus S	Stimulus I	Expressio	Threshold	Expression Inde	Expression Index	Expression Ons	Expression Offset	Expression Duratic	Truncation Type	Duration Classification	Time Threshold	Device
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Engageme	50	1	1	463	463	0	Whole	Short	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Positive Vi	50	1	1	463	463	0	Whole	Short	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Cheek Rai	20	1	1	463	930	467	Whole	Short	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Smile	20	1	1	463	1007	544	Whole	Short	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Eye Closu	20	1	1	463	1344	881	Whole	Short	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Neutral	50	1	1	463	3711	3248	Whole	Long	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Head Turr	-10 | 10		1	1	463	5312	4849	Whole	Long	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Head Not	-10 | 10		1	1	463	5745	5282	Whole	Long	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Attention	50	1	1	463	6256	5793	Whole	Long	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Head Lear	100	1	1	463	9488	9025	Whole	Long	1000	Affectiva AFFDEX
Media (Skurvo	Anonymot	MALE	0	Default	Stryhns - F	432.7113	35047.02	Neutral Vs	-50 | 50		1	1	666	4815	4149	Whole	Long	1000	Affectiva AFFDEX

The facial expression table export

Sensor Data

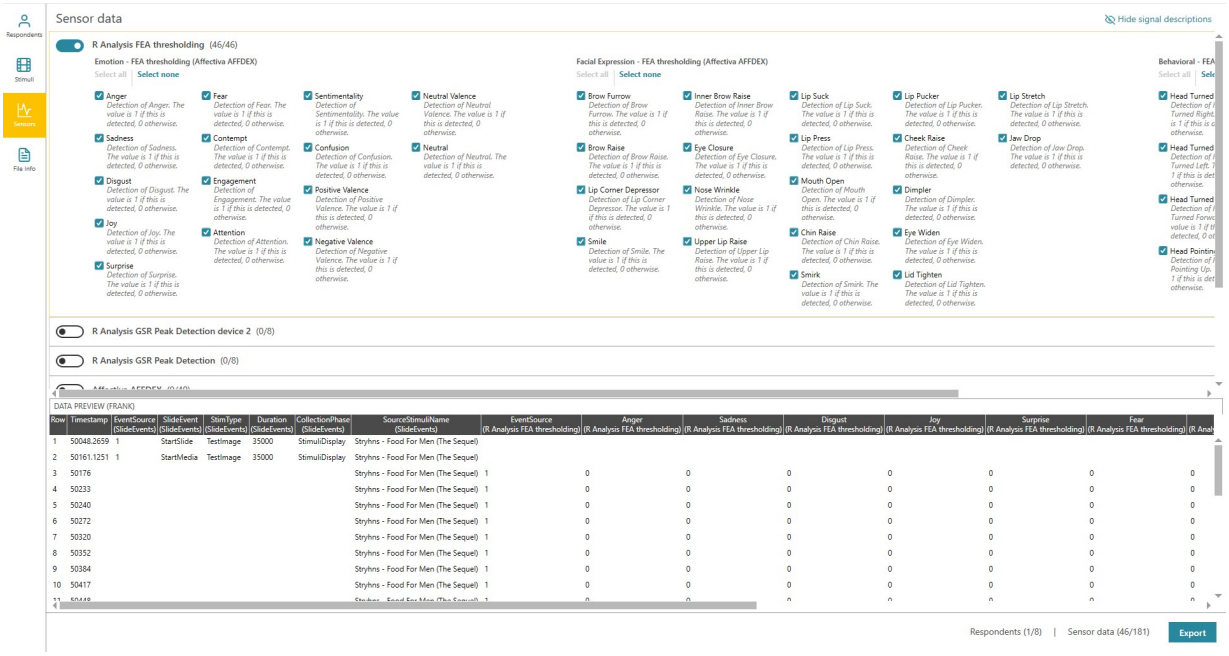
Under *Built in processing* > *Sensor Data*, you can choose to export time series data.



Access the Sensor Data export through *Export* > *Built in Processing* > *Sensor Data* > *Export*

Here, you can choose to export the binarized signals.

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UI to include the binarized facial expression signals in the Sensor Data export

Next steps

If you open the exports in Excel, Pivot tables are a helpful tool to compare the results from different conditions

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